

IoT Based Agricultural Monitoring and Alert System

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Abstract—: *The goal of this project is to implement an Internet of Things (IoT)-based Agricultural Monitoring & Alert System (AMAS) that will integrate multiple sensors to continuously monitor agricultural parameters, such as soil moisture levels, soil temperature, water levels, irrigation pump operation status, and water quality, to improve precision farming practices. Enabling farmers to make well-informed decisions, maximize resources, and enhance crop output in real time is the primary objective of the AMAS system. The challenges of using IoT for agricultural monitoring, include issues with adoption among farmers, dependable technology, efficient data administration, and connectivity. Small-scale farmers' financial hardship and rural areas' restricted access to technology. Thus, we plan to enhance productivity, optimize yield, and implement remote monitoring.*

Keywords—Arduino-ESP32-LM35-Ultrasonics-Moisture sensor-Blynk-connecting wires-LED bulbs-TDS-Pump motor-Battery.

1. INTRODUCTION

The Internet of Things (IoT) has had a profound impact on agriculture and is causing a transformative wave of innovation. An intelligent and streamlined method of crop management in the farming sector is introduced by the Internet of Things (IoT)-based Agricultural Monitoring and Alert System. Thanks to smart sensors and connectivity, this system gives farmers access to real-time data on important parameters like soil temperature, water level, pump status, moisture content, and water quality. Farmers can make well-informed decisions, optimize resource utilization, and receive timely alerts thanks to the continuous data collection provided by smart sensors. The system makes agriculture more productive and efficient, resulting in a farming ecosystem that is more resilient and adaptable. By enabling farmers to handle the complexities of crop management with accuracy and foresight, this represents a major advancement in precision agriculture.

2. LITERATURE REVIEW

[1] To protect food crops from untamed animals, create an Internet of Things (IoT)-based electric fence notification system that enables farmers to remotely check the battery level and fence status. [2] Provide an intelligent farming system that can automatically monitor soil conditions, regulate irrigation, suggest

crops, and control pests to increase crop yields and farming efficiency for Indian farmers. [3] Establish a precision agriculture sensing network with IoT technology, emphasizing crop quality maximization and environmental impact minimization. [4] To improve productivity and address the challenges posed by climate change, implement an Internet of Things (IoT)-based smart farming technique that automates farm monitoring, crop control, and data collection. [5] In order to minimize human intervention, increase the efficiency of irrigation systems, and automate crop irrigation for sustainable agriculture, soil moisture content should be assessed. [6] Examine Internet of Things (IoT)-based approaches to modernizing agriculture, such as data analytics, network architecture, and smartphone apps, while taking security and policy issues into consideration. [7] In India, use IoT to automate irrigation systems, saving labor, extending the life of pump motors, and guaranteeing effective water use for crop cultivation. [8] Give a study on the Internet of Things (IoT) and farm monitoring, emphasizing the development of farming technology and the potential advantages of smart farming for raising yield and supplying future food demands. [9] In order to increase crop health and yield, suggest an Internet of Things (IoT)-based precision agriculture monitoring system to collect information on plant development, diseases, soil properties, and irrigation requirements. [10] Create a cost-effective IoT-powered smart farming system with temperature and moisture sensors in the soil to track crop parameters and raise agricultural output. [11] To help farmers remotely monitor crops, optimize growth conditions, and make well-informed decisions for increased yields and profitability, introduce an Internet of Things (IoT)-based smart agriculture monitoring system. [12] Use IoT and machine learning to implement a precision agriculture system that will measure environmental parameters, maximize resource use, and increase crop yields. [13] Examine how wireless networks and sensor technology are integrated in the Internet of Things (IoT) for agriculture, with an emphasis on real-time data collection and decision support to enhance agricultural practices. [14] Create a smart irrigation system that uses machine learning and the Internet of Things to anticipate crop water requirements based on environmental conditions. This will help farmers plan their crops and manage water more effectively. [15] Provide an IoT-based web application that allows farmers to remotely access

information about temperature, humidity, and soil moisture for effective irrigation management and other farm-related tasks.[16] presenting the case for automation in agriculture to address issues facing the industry, especially in India, and emphasizing the advantages of an autonomous irrigation system in terms of increased output and decreased manual labor.[17] Introducing a smart farming model based on the Internet of Things that employs clustering techniques to automate data collection, manage crop conditions, and improve productivity in the face of unfavorable conditions. [18] offering an Internet of Things (IoT) system that uses wireless sensor networks and cloud computing to monitor and manage agricultural irrigation in real time, maximizing water efficiency and reducing the need for human intervention. [19] talking about how the Internet of Things is changing agriculture by making it possible to collect data on environmental factors, monitor crops online, and manage farms remotely, all of which will increase productivity and cut costs.[20] presenting a synopsis of current advancements and technologies in Internet of Things (IoT)-based smart farming, talking about commercial products, and suggesting future paths for research..[21] Using real-time environmental monitoring to optimize water consumption, stop plant theft, and improve crop protection, an intelligent monitoring and automated irrigation system can be developed using the Internet of Things.[22] ntroducing a framework for utilizing IoT in agriculture with the goal of increasing sustainability and productivity. This includes automating irrigation, detecting pests, and boosting agricultural growth. [23] Putting together a smart agriculture system that makes use of wireless sensor networks, IoT, and smartphone apps to track weather patterns, identify irregularities, and increase crop yield all while keeping costs down. [24] Propose an automated farm monitoring system for Bangladesh that focuses on the important role of agriculture in managing the rapid growth of the world's population and the increasing demand for food. The system uses a cloud-based IoT solution with multiple NodeMCU (ESP8266) microcontrollers to monitor and manage critical farm areas. Continuous monitoring of farm components allows the central control unit to analyze data and assign specific tasks to individual microcontrollers. [25] Recommending the use of a cloud-based IoT solution for an automated farm control system to monitor and manage key farm areas in order to address issues with agricultural productivity. [26] creating an inexpensive IoT-based pond water quality monitoring system that can continuously monitor important aquaponics system parameters.

[27] nvestigating the feasibility of an Arduino-based sensor system for water quality monitoring with the goal of increasing the precision and effectiveness of water quality assessment..[28] Introducing an Internet of Things (IoT)-based smart water quality monitoring system that will address worries about water pollution and safety by continuously monitoring water parameters, detecting contamination, and ensuring a supply of pure water..[29] Describing an Internet of Things (IoT)-based water monitoring system that will monitor water quality, control pipeline networks, find leaks, and alert users to deviations in order to increase water management's effectiveness and safety. [30] Creating an affordable, real-time water quality control system with Internet of Things (IoT) capabilities to track multiple water quality parameters and guarantee security via web-based data visualization.

III. PROPOSED WORK

A soil moisture sensor is used to measure and provide information about the amount of water present in the soil. An ultrasonic sensor is used to tell how much water is present in the area, if the water level is low or high it will help to monitor the water level for the area. A pump status sensor is used to provide information about the operational status of the pump whether it is ON/OFF. Total Dissolved Solids (TDS) sensors measure the amounts of TDS in water, which can be used to determine the

quality of the water. A temperature sensor is used to measure the temperature in the soil. It gives the values in the form of degrees like Celsius or Fahrenheit. These soil moisture, water level, pump status, and TDS sensors are connected to analog pins and the temperature sensor is connected to the power supply pin. A Wi-Fi module is used to connect devices such as phones, laptops, etc... For altering the system, we use the Buzzer. To enable smooth communication between the system and farmers, this notification mechanism can be expanded to include SMS, updates for mobile apps, or other channels.

IV. SOFTWARE AND HARDWARE WORK

Arduino:

An open-source hardware and software platform called Arduino is used to create interactive and digital products. It is made up of an integrated development environment (IDE) that enables users to write and upload code to a microcontroller board.

Blynk

One well-liked platform for creating Internet of Things (IoT) applications is Blynk. It offers an easy method for developing mobile apps that manage and keep an eye on Internet of Things devices. Blynk is especially well-known for its intuitive interface and capacity to rapidly prototype Internet of Things projects without requiring a lot of coding.

A succinct description of Blynk's functionality within the IoT:

- **Blynk App:** The Blynk app is accessible on Android and iOS platforms. It enables users to design their own interface using widgets such as graphs, displays, sliders, and buttons. For IoT devices, this interface functions as a remote control.
- **Blynk Cloud Server:** Blynk makes use of its cloud server to make it easier for the Blynk app and IoT devices to communicate with one another. By transmitting data and commands between the application and the hardware, the server serves as a bridge.
- **Hardware Supported by Blynk:** A variety of microcontrollers and development boards, such as Arduino, Raspberry Pi, ESP8266, ESP32, and others, are supported by Blynk. It works with several communication protocols, including Ethernet, GSM, Bluetooth, and Wi-Fi.
- **Blynk Libraries:** You must include the Blynk libraries in your programming environment to integrate Blynk with your IoT hardware. These libraries offer features and techniques for handling communication and connecting your device to the Blynk cloud.
- **Authentication Token:** A distinct key that connects the hardware device to the Blynk app project is assigned to every Blynk project. Secure communication between the app and the device is ensured by this token.

Hardware Specification

Node MCU

A development board built around the ESP32 microcontroller is called the NodeMCUESP32. It has several uses and is well-liked in the embedded systems and Internet of Things domains.

Temperature Sensor

- A popular temperature sensor for determining the outside temperature is the LM35.
- It outputs a linear voltage that is precisely

proportional to the temperature in Celsius.

- The LM35 can be buried in the ground to monitor and gauge the temperature of the surrounding soil in the context of soil temperature sensing.

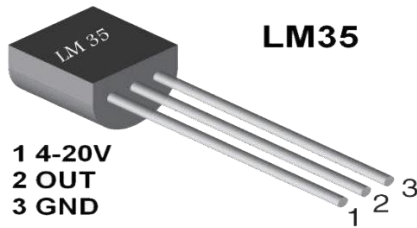


Fig A. Temperature sensor

Ultrasonic Sensor

- A common tool for measuring distance and determining water levels in tanks is the HC-SR04 Ultrasonic Sensor.
- The sensor uses ultrasonic waves to measure the time it takes for echoes to return in order to determine how far away it is from the water's surface.
- Applications including water level monitoring in tanks and reservoirs are common because of the accuracy with which the water level can be determined thanks to this data.

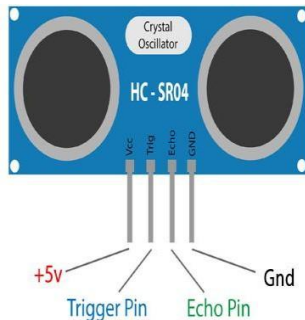


Fig B. Ultrasonic Sensor

Moisture sensor

- A moisture sensor is a tool used to determine the amount of moisture or humidity present in a specific material, most commonly soil in agricultural settings.
- Moisture sensors are useful in gardening and agriculture because they help determine how much water is in the soil, which helps with irrigation decisions

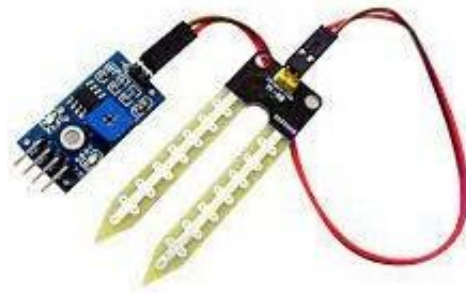


Fig C. Moisture sensor

LED Indicator

- A tiny light-emitting diode (LED) used to visually represent information or status is called an LED indicator.
- Turned on, it emits light, acting as a visual signal or display.
- LED indicators are frequently used in a variety of systems and devices to indicate different states, such as connectivity, power on/off, or the occurrence of events.
- Through the LED's illumination or blinking, they offer a quick and easy way for users to determine whether a system or device is operating normally.



Fig D. LED Bulbs

TDS sensor

- A Total Dissolved Solids (TDS) sensor is a tool used to quantify the number of dissolved materials in a liquid, including salts, minerals, and other contaminants.
- TDS sensors are frequently used in agriculture and water quality monitoring to evaluate the general purity or contamination of water.
- The sensor yields a numerical value that represents the total amount of dissolved solids in the liquid under test, usually expressed in parts per million (ppm) or milligrams per liter (mg/L).
- For ensuring water quality in a variety of applications, such as environmental monitoring and irrigation management, this information is helpful.



Fig E. TDS Sensor

Hardware setup methodology

Physical connection is established by connecting the Soil Moisture sensor, TDS, LM35 and HC-SR04 sensor to the NodeMCU board. Status indicator LEDs are strategically connected to the NodeMCU development board. The field motor is connected to the Battery 3v in the tank for the watering when the water level low in the field. The Tank motor is connected to the Battery 3v for the when the water level is less in the tank it will automatically fill the water the tank. The Buzzer is connected to the NodeMCU board for the alerting. When the field motor and tank motor is ON/OFF it will alert for a certain amount of time. The LEDs indicator is used for the alert purposes here, when the field motor is ON and it will be pouring the water to the field at that time Beep sound is come from buzzer with the LEDs indicators. For the notification purposes we use the Blynk application. Blynk will also display the live values of the soil moisture sensor, TDS, LM35 and HC-SR04 sensor. The LED bulb also glow to show the current status of process.

Programming Methodology

- Writing code to collect data from sensors, such as soil moisture, temperature, water levels, and water quality, is a necessary part of programming IoT with NodeMCU ESP32.
- After that, this data is sent to an Internet of Things platform for analysis and visualization.
- In addition, the code contains logic to regulate irrigation pumps and other devices according to sensor readings.
- The Arduino IDE is used to program the NodeMCU ESP32, and libraries are utilized to interface with various sensors.

Blynk Application Connectivity Methodology

Blynk is an Internet of Things (IoT) platform that enables connectivity between hardware devices (such as microcontrollers or development boards) and mobile applications.

Work Flow Diagram:

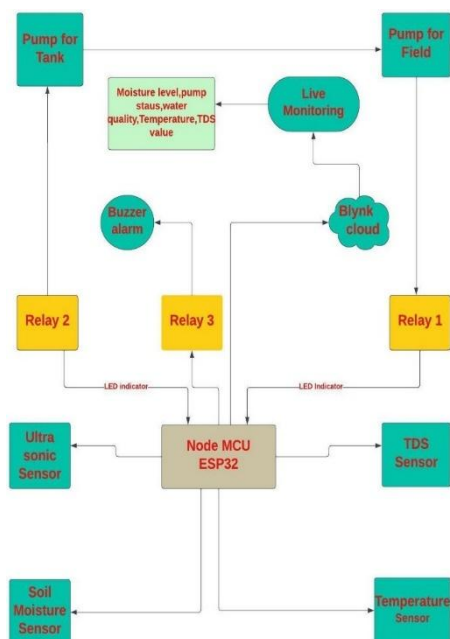


Fig I. Work Flow

VI.RESULTS AND DISCUSS

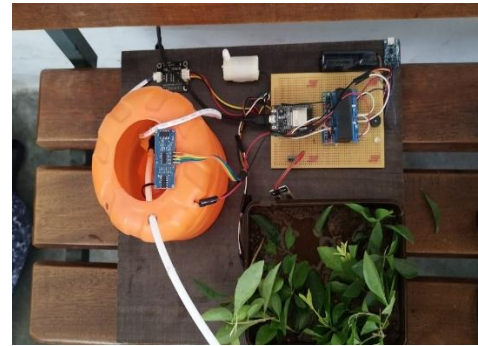


Fig J. Model setup

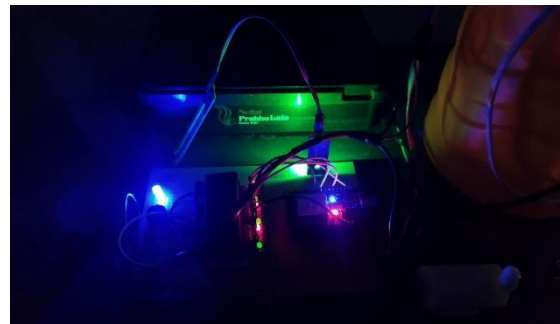


Fig K. Model Working

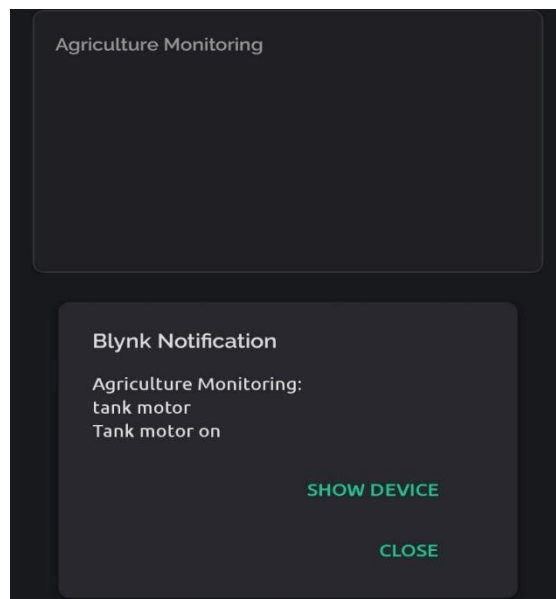


Fig L. Agricultural Monitoring Interface

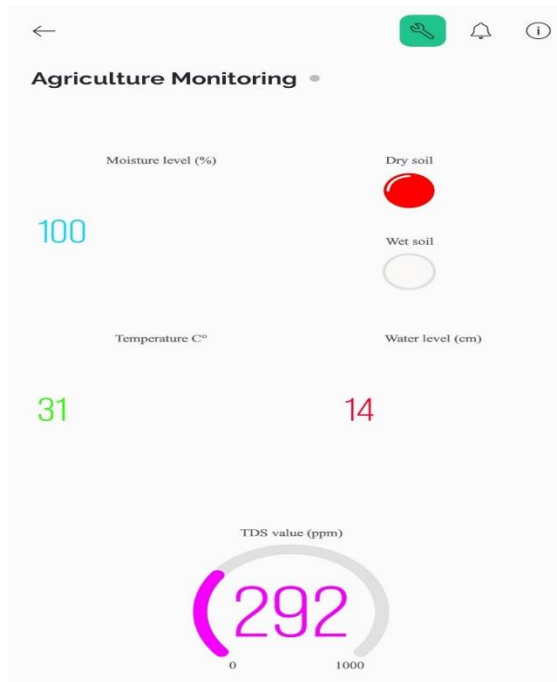


Fig M. Reading value



Fig N. Graph Reading

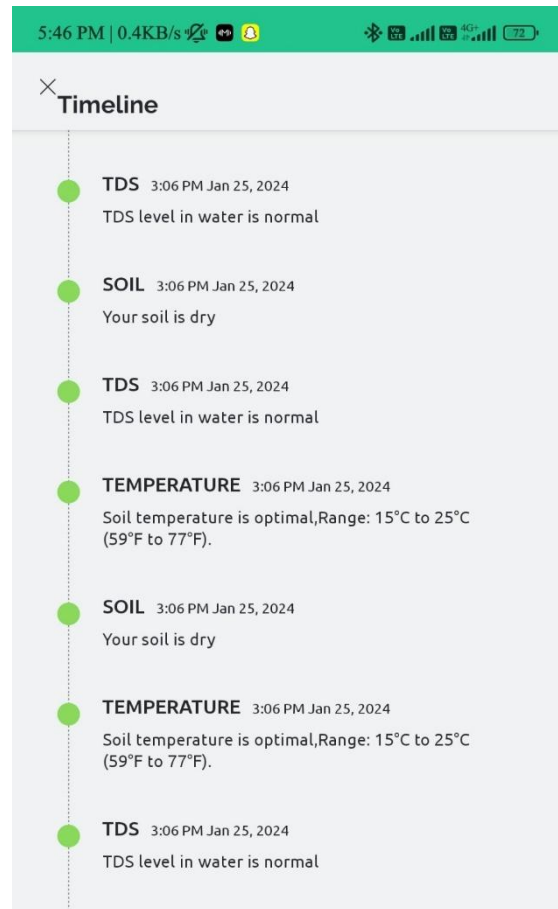


Fig O. Notification Reading

A soil moisture sensor to measure soil water content and optimize irrigation, an ultrasonic sensor to track water levels and notify farmers of deviations, a pump status sensor to monitor irrigation system operation, Total Dissolved Solids (TDS) sensors to evaluate water quality, and a temperature sensor to monitor soil temperature and evaluate crop health are all included in the integrated agricultural monitoring system. The system makes use of a buzzer for on-site alerts or alarms and a Blynk module for real-time remote monitoring and control. By expanding communication options to include SMS or mobile app updates, the notification mechanism makes it easier for farmers to access vital information on the platforms of their choice and encourages effective and knowledgeable agricultural practices. By combining these components, farmers can make informed decisions about irrigation, monitor water quality, and ensure the overall health of their crops. Additionally, the incorporation of diverse notification channels enhances the system's versatility, ensuring that farmers receive timely information in a format that suits their preferences. This comprehensive approach contributes to more efficient and sustainable agricultural practices.

VII.CONCLUSION

In a nutshell, the smart farming system employs a variety of sensors to assist farmers in providing better care for their crops. To determine when to water the plants, a

sensor measures the amount of water in the soil. Another monitors the levels of the water and alerts the farmers if something is not right. An additional sensor for the pump is included to ensure proper operation. Moreover, sensors are used to determine whether the soil temperature is appropriate for the crops and whether the water is clean. All of this is linked to a system that can send notifications remotely via the internet or on-site using a buzzer. In this manner, farmers can more easily obtain the information they require to improve the efficiency of their farms.

REFERENCE:

- [1] A. S. J. Ali et al., "Towards IoT-based Notification System for Agriculture Electric Fence," in *2020 IEEE Student Conference on Research and Development (SCoReD)*, Batu Pahat, Malaysia: IEEE, Sep. 2020, pp. 269–273. doi: 10.1109/SCoReD50371.2020.9250958.
- [2] N. Ananthi, J. Divya, M. Divya, and V. Janani, "IoT based smart soil monitoring system for agricultural production," in *2017 IEEE Technological Innovations in ICT for Agriculture and Rural Development (TIAR)*, Chennai: IEEE, Apr. 2017, pp. 209–214. doi: 10.1109/TIAR.2017.8273717.
- [3] A. A. Araby et al., "Smart IoT Monitoring System for Agriculture with Predictive Analysis," in *2019 8th International Conference on Modern Circuits and Systems Technologies (MOCASST)*, Thessaloniki, Greece: IEEE, May 2019, pp. 1–4. doi: 10.1109/MOCASST.2019.8741794.
- [4] M. Ashifuddin Mondal and Z. Rehena, "IoT Based Intelligent Agriculture Field Monitoring System," in *2018 8th International Conference on Cloud Computing, Data Science & Engineering (Confluence)*, Noida, India: IEEE, Jan. 2018, pp. 625–629. doi: 10.1109/CONFLUENCE.2018.8442535.
- [5] J. Boobalan, V. Jacintha, J. Nagarajan, K. Thangayogesh, and S. Tamilarasu, "An IOT Based Agriculture Monitoring System," in *2018 International Conference on Communication and Signal Processing (ICCSP)*, Chennai: IEEE, Apr. 2018, pp. 0594–0598. doi: 10.1109/ICCSP.2018.8524490.
- [6] M. S. Farooq, S. Riaz, A. Abid, K. Abid, and M. A. Naeem, "A Survey on the Role of IoT in Agriculture for the Implementation of Smart Farming," *IEEE Access*, vol. 7, pp. 156237–156271, 2019, doi: 10.1109/ACCESS.2019.2949703.
- [7] S. Gnanavel, M. Sreekrishna, N. Durai Murugan, M. Jaeyalakshmi, and S. Loksharan, "The Smart IoT based Automated Irrigation System using Arduino UNO and Soil Moisture Sensor," in *2022 4th International Conference on Smart Systems and Inventive Technology (ICSSIT)*, Tirunelveli, India: IEEE, Jan. 2022, pp. 188–191. doi: 10.1109/ICSSIT53264.2022.9716368.
- [8] V. Lohchab, M. Kumar, G. Suryan, V. Gautam, and R. K. Das, "A Review of IoT based Smart Farm Monitoring," in *2018 Second International Conference on Inventive Communication and Computational Technologies (ICICCT)*, Coimbatore: IEEE, Apr. 2018, pp. 1620–1625. doi: 10.1109/ICICCT.2018.8473337.
- [9] I. M. Marcu, G. Suciu, C. M. Balaceanu, and A. Banaru, "IoT based System for Smart Agriculture," in *2019 11th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)*, Pitesti, Romania: IEEE, Jun. 2019, pp. 1–4. doi: 10.1109/ECAI46879.2019.9041952.
- [10] D. Mishra, T. Pande, K. K. Agrawal, A. Abbas, A. K. Pandey, and R. S. Yadav, "Smart agriculture system using IoT," in *Proceedings of the Third International Conference on Advanced Informatics for Computing Research*, Shimla India: ACM, Jun. 2019, pp. 1–7. doi: 10.1145/3339311.3339350.
- [11] B. Mushtaq, H. Ali, Engr Rizwan Latif, and M. Ammar, "IOT Based Smart Agriculture Monitoring System," 2023, doi: 10.13140/RG.2.2.14040.47367.
- [12] G. S. Nagaraja, A. B. Soppimath, T. Soumya, and A. Abhinith, "IoT Based Smart Agriculture Management System," in *2019 4th International Conference on Computational Systems and Information Technology for Sustainable Solution (CSITSS)*, Bengaluru, India: IEEE, Dec. 2019, pp. 1–5. doi: 10.1109/CSITSS47250.2019.9031025.
- [13] K. A. Patil and N. R. Kale, "A model for smart agriculture using IoT," in *2016 International Conference on Global Trends in Signal Processing, Information Computing and Communication (ICGTSPICC)*, Jalgaon, India: IEEE, Dec. 2016, pp. 543–545. doi: 10.1109/ICGTSPICC.2016.7955360.
- [14] K. S. Pratyush Reddy, Y. M. Roopa, K. Rajeev L.N., and N. S. Nandan, "IoT based Smart Agriculture using Machine Learning," in *2020 Second International Conference on Inventive Research in Computing Applications (ICIRCA)*, Coimbatore, India: IEEE, Jul. 2020, pp. 130–134. doi: 10.1109/ICIRCA48905.2020.9183373.
- [15] K. Raviteja and M. Supriya, "IoT-Based Agriculture Monitoring System," in *Data Engineering and Communication Technology*, vol. 1079, K. S. Raju, R. Senkerik, S. P. Lanka, and V. Rajagopal, Eds., in *Advances in Intelligent Systems and Computing*, vol. 1079., Singapore: Springer Singapore, 2020, pp. 473–483. doi: 10.1007/978-981-15-1097-7_40.
- [16] N. Revathy, T. Guhan, S. Nandhini, S. Ramadevi, and R. Dhipthi, "IOT based Agriculture Monitoring System using Arduino UNO," in *2022 International Conference on Computer Communication and Informatics (ICCCI)*, Coimbatore, India: IEEE, Jan. 2022, pp. 01–05. doi: 10.1109/ICCCI54379.2022.9740910.
- [17] M.K. Saini and R. K. Saini, "Agriculture monitoring and prediction using Internet of Things (IoT)," in *2020 Sixth International Conference on Parallel, Distributed and Grid Computing (PDGC)*, Waknaghat, India: IEEE, Nov. 2020, pp. 53–56. doi: 10.1109/PDGC50313.2020.9315836.
- [18] S. B. Saraf and D. H. Gawali, "IoT based smart irrigation monitoring and controlling system," in *2017 2nd IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)*, Bangalore: IEEE, May 2017, pp. 815–819.

doi: 10.1109/RTEICT.2017.8256711.

HTC53172.2021.9641630.

[19] D. K. Sreekantha and Kavya A.M., "Agricultural crop monitoring using IOT - a study," in 2017 11th International Conference on Intelligent Systems and Control (ISCO), Coimbatore, India: IEEE, Jan. 2017, pp. 134–139. doi: 10.1109/ISCO.2017.7855968.

[20] A. Srivastava and D. K. Das, "A Comprehensive Review on the Application of Internet of Thing (IoT) in Smart Agriculture," *Wireless Pers Commun*, vol. 122, no. 2, pp. 1807–1837, Jan. 2022, doi: 10.1007/s11277-021-08970-7.

[21] A. F. Suhaimi et al., "IoT Based Smart Agriculture Monitoring, Automation and Intrusion Detection System," *J. Phys.: Conf. Ser.*, vol. 1962, no. 1, p. 012016, Jul. 2021, doi: 10.1088/1742-6596/1962/1/012016.

[22] Md. F. I. Suny et al., "Smart Agricultural System Using IoT," in *Intelligent Sustainable Systems*, vol. 333, A. K. Nagar, D. S. Jat, G. Marín-Raventós, and D. K. Mishra, Eds., in *Lecture Notes in Networks and Systems*, vol. 333, Singapore: Springer Nature Singapore, 2022, pp. 73–82. doi: 10.1007/978-981-16-6309-3_8.

[23] G. Sushanth and S. Sujatha, "IOT Based Smart Agriculture System," in 2018 International Conference on Wireless Communications, Signal Processing and Networking (WiSPNET), Chennai: IEEE, Mar. 2018, pp. 1–4. doi: 10.1109/WiSPNET.2018.8538702.

[24] M. S. Tajwar Haque, K. Abdur Rouf, Z. A. Khan, A. Emran, and Md. S. Rahman Zishan, "Design and Implementation of an IoT based Automated Agricultural Monitoring and Control System," in 2019 International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST), Dhaka, Bangladesh: IEEE, Jan. 2019, pp. 13–16. doi: 10.1109/ICREST.2019.8644212.

[25] J. Yang, A. Sharma, and R. Kumar, "IoT-Based Framework for Smart Agriculture," *International Journal of Agricultural and Environmental Information Systems*, vol. 12, no. 2, pp. 1–14, Apr. 2021, doi: 10.4018/IJAEIS.20210401.oal.

[26] N. D. Susanti, D. Sagita, I. F. Apriyanto, C. E. W. Anggara, D. A. Darmajana, and A. Rahayuningtyas, "Design and Implementation of Water Quality Monitoring System (Temperature, pH, TDS) in Aquaculture Using IoT at Low Cost," presented at the 6th International Conference of Food, Agriculture, and Natural Resource (IC-FANRES 2021), Tangerang, Indonesia, 2022. doi: 10.2991/absr.k.220101.002.

[27] W. Hong et al., "Water Quality Monitoring with Arduino Based Sensors," *Environments*, vol. 8, no. 1, p. 6, Jan. 2021, doi: 10.3390/environments8010006.

[28] V. Lakshmikantha, A. Hiriyannagowda, A. Manjunath, A. Patted, J. Basavaiah, and A. A. Anthony, "IoT based smart water quality monitoring system," *Global Transitions Proceedings*, vol. 2, no. 2, pp. 181–186, Nov. 2021, doi: 10.1016/j.gltp.2021.08.062.

[29] G. N. Satya Sai, R. Sudheer, K. S. Manikanta, S. G. Arjula, B. N. Rao, and D. V. Sai Maneeswar Mutyala, "IoT based Water Quality Monitoring System," in 2021 IEEE 9th Region 10 Humanitarian Technology Conference (R10-HTC), Bangalore, India: IEEE, Sep. 2021, pp. 01–06. doi: 10.1109/R10-

[30] A. H. Abdulwahid, "IoT Based Water Quality Monitoring System for Rural Areas," in 2020 9th International Conference on Renewable Energy Research and Application (ICRERA), Glasgow, United Kingdom: IEEE, Sep. 2020, pp. 279–282. doi: 10.1109/ICRERA49962.2020.9242798.